

LinHT RFIC - SG13CMOS5L | 130–520 MHz RF Frontend IC

System Architecture - Short Design Reference

1. Top-Level Architecture

The LinHT RFIC is a highly integrated half-duplex RF transceiver intended to provide continuous operation over approximately **130–520 MHz**. The architecture is organized into five major functional groups: **RF and frequency generation, receiver, transmitter, control and calibration, and power/I/O**. This partitioning provides clear design ownership while allowing the individual RF, analog, mixed-signal, and digital macros to be independently designed and verified before full-chip integration.



Figure 1. LinHT RFIC — Architecture Overview

1.1 RF and Frequency Generation

The frequency-generation chain begins with the **XO/reference clock**, nominally around 32 MHz, with support for approximately 26–40 MHz reference sources. This reference drives the **fractional-N PLL**, which generates the accurate frequency control required for channel selection.

The PLL controls an integrated **LC VCO operating approximately from 2.08 to 4.16 GHz**. Rather than operating the VCO directly at the final RF frequency, the architecture uses a high-frequency oscillator followed by frequency division. The LO divider provides the required RF bands using approximately **÷8 and ÷16 division paths**, covering the overall 130–520 MHz operating range.

A final quadrature generation stage produces **I and Q LO signals with approximately 90° phase separation**. These quadrature signals are distributed to both the receive and transmit mixers. Sharing a common synthesizer and LO distribution network ensures coherent RX and TX carrier generation while reducing duplication of high-frequency circuitry.

1.2 Receiver

The receiver converts a weak RF signal from the antenna interface into digital I/Q samples. The signal first enters the **RX input** and is amplified by the **low-noise amplifier (LNA)**. Because the LNA is the first active stage, its noise figure and gain strongly influence overall receiver sensitivity.

The amplified RF signal is applied to the **quadrature RX mixer**, where the LO_I and LO_Q signals translate the selected RF channel to zero-IF or low-IF baseband.

The resulting I and Q signals are processed by the analog baseband chain consisting of a **TIA, programmable low-pass filter, and PGA**. These stages provide current-to-voltage conversion, channel filtering, and programmable gain before analog-to-digital conversion.

Two **continuous-time sigma-delta ADCs**, one for I and one for Q, digitize the baseband signals. The resulting digital streams are decimated, filtered, and corrected by the **RX DSP** before being transferred to the external host through I²S.

The complete RX path can therefore be summarized as:

RF Input → LNA → IQ Mixer → TIA/LPF/PGA → I/Q ΣΔ ADC → RX DSP → I²S

1.3 Transmitter

The transmitter performs the inverse operation. Digital I/Q samples arrive through the I²S interface and enter the **TX DSP**, where interpolation, digital filtering, and sigma-delta processing are performed.

The processed I/Q data drives the **FIR DACs**, which reconstruct the analog baseband waveform. Analog low-pass filters then suppress DAC images and out-of-band sigma-delta quantization noise.

The filtered I/Q signals enter the **quadrature TX mixer**, where the common LO system translates them to the selected RF carrier frequency. A programmable **RF output driver** then provides the required output signal to the differential RF interface.

The complete TX path is:

I²S → TX DSP → FIR DAC I/Q → TX LPF → IQ Up-Mixer → RF Driver → RF Output

1.4 Supporting Infrastructure

The remaining chip functions support these main signal paths. **SPI registers and the control sequencer** configure operating frequency, bandwidth, gain, modes, and calibration. The calibration subsystem compensates VCO, RC, I/Q, and DC variations. Finally, the **bias, POR, power-domain, pad, and ESD circuitry** provides the electrical infrastructure required for reliable chip operation.

2. Signal-Path Architecture

The second diagram emphasizes how the **frequency synthesizer, receiver, and transmitter interact**. The key architectural concept is that a single frequency-generation subsystem produces the quadrature LO required by both signal directions.

2.1 Reference, PLL and VCO

Frequency generation begins with the nominal **32 MHz reference**. The fractional-N PLL compares this reference against a divided version of the VCO output and continuously adjusts the VCO control voltage until the required frequency relationship is achieved.

The fractional-N capability provides fine frequency resolution without requiring an extremely low reference frequency. A **20-bit MASH 1-1-1 sigma-delta modulator** controls the fractional divider sequence, allowing the average feedback division ratio to assume fractional values.

The PLL drives an **LC VCO covering approximately 2.08–4.16 GHz**. Operating the VCO at several times the desired RF frequency allows the architecture to use divider stages to obtain the required RF range while simultaneously helping with quadrature generation.

The LO chain produces approximately:

- **260–520 MHz through the higher-frequency division path**
- **130–260 MHz through the lower-frequency division path**

Together, these provide continuous LO coverage over the intended **130–520 MHz RF range**.

The final divider stage produces quadrature **LO_I and LO_Q signals**, which are routed to both the RX and TX mixers.

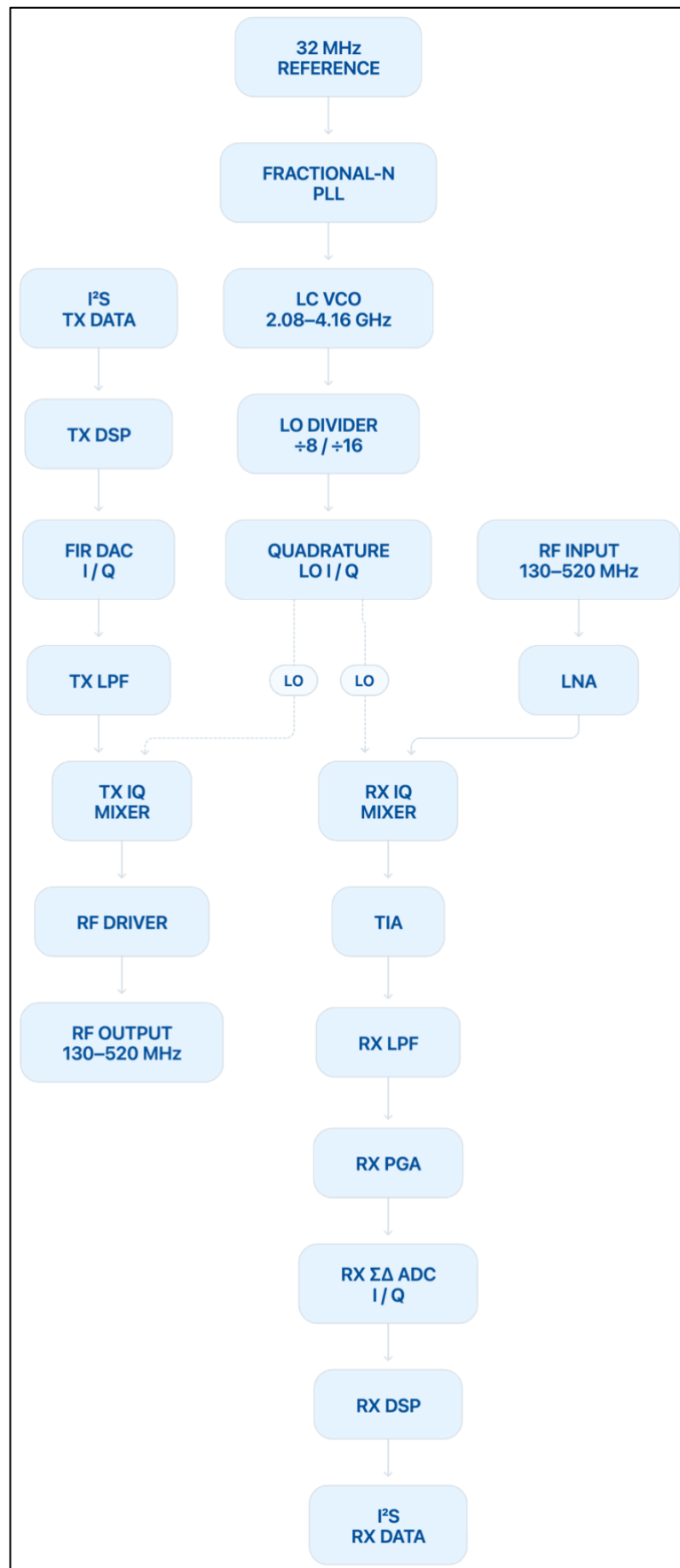


Figure 2. RX/TX Signal Flow and Frequency Generation

2.2 Receive Signal Flow

The received RF signal first passes through the **LNA**. The primary purpose of this stage is to provide sufficient gain while adding minimal noise. Consequently, LNA noise figure, gain, linearity, input matching, and power consumption are major receiver design parameters.

The signal then enters the **IQ down-mixer**. Mixing the RF signal with quadrature LO phases creates separate I and Q baseband components that retain the amplitude and phase information required by the SDR.

After mixing, the signal passes through the **TIA, RX LPF, and PGA**. The TIA converts mixer output current into voltage. The LPF limits the receive bandwidth and removes unwanted mixer products, while the PGA adjusts signal amplitude to use the available ADC input range effectively.

Separate I and Q **third-order continuous-time sigma-delta ADCs** convert the analog baseband signals into digital streams. Digital processing then performs decimation, filtering, DC removal, and IQ correction before delivering 16-bit I/Q samples through I²S.

This combination of analog filtering and digital processing allows the receiver to trade bandwidth, sample rate, noise, and dynamic range depending on operating mode.

2.3 Transmit Signal Flow

The TX chain begins with digital I/Q samples from the host. The **TX DSP** performs interpolation and digital filtering before applying sigma-delta processing.

The **FIR DAC I/Q pair** converts the processed digital information into analog baseband signals. The following TX LPFs remove high-frequency DAC images and shaped quantization noise before the signal reaches the RF mixer.

The **IQ up-mixer** combines the analog I and Q signals with LO_I and LO_Q to translate the complex baseband waveform to the selected RF carrier.

Finally, the **RF driver** amplifies the modulated RF waveform and drives the differential RF output.

2.4 Importance of the Shared LO

The shared LO is one of the most important architectural elements of the chip. PLL phase noise, VCO noise, divider jitter, LO duty-cycle error, and quadrature phase mismatch can all influence both RX and TX performance.

For this reason, the synthesizer cannot be designed independently of the RF signal chain. Its phase-noise and spur requirements must be derived from receiver selectivity and transmitter spectral requirements.

Similarly, the LO distribution network must provide sufficient mixer drive while minimizing coupling into the LNA, baseband, ADC, DAC, and external RF pins.

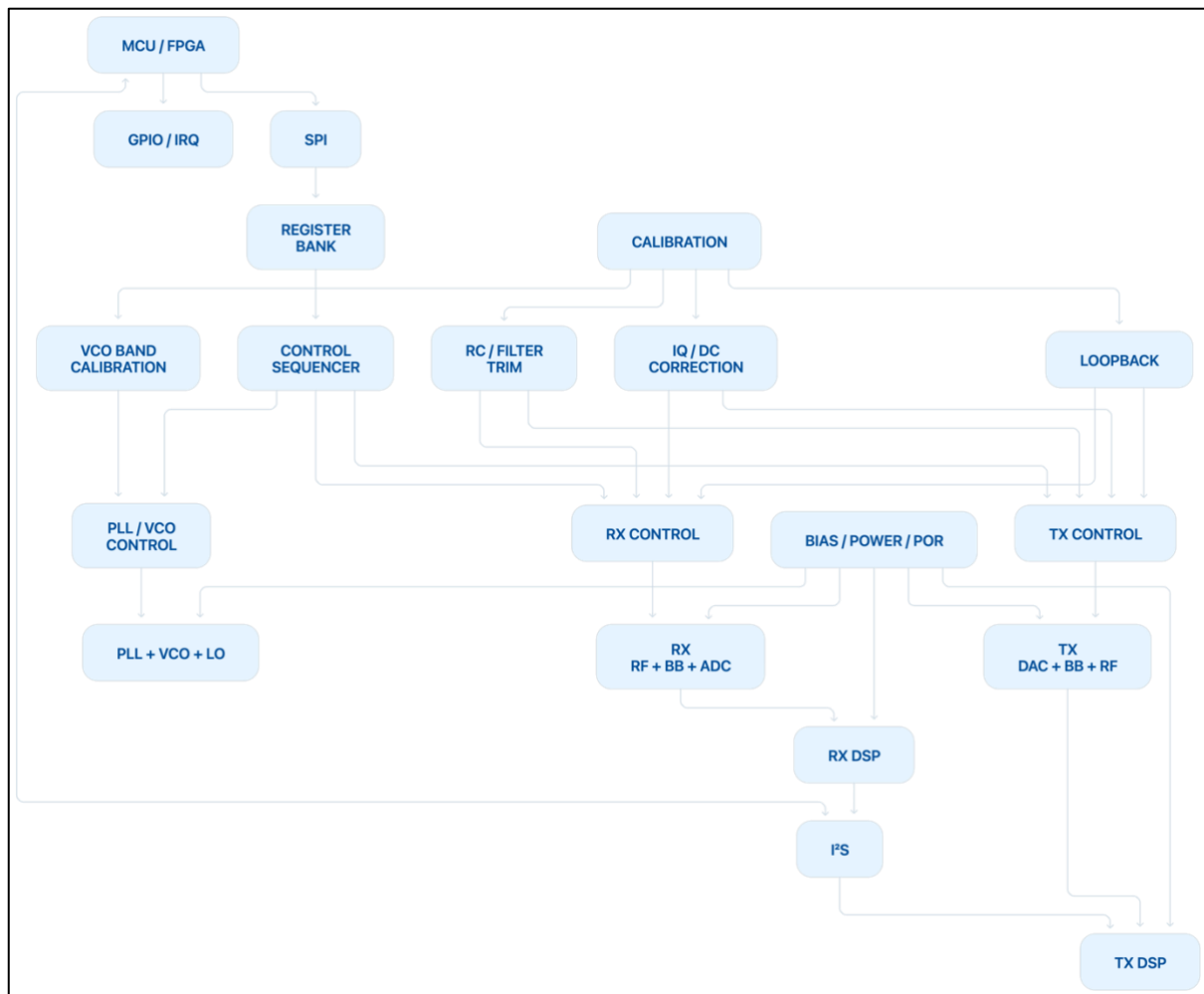


Figure 3. Digital Control, Calibration and Power Architecture

3. Control and Calibration Architecture

The third diagram describes the supervisory infrastructure surrounding the RF and mixed-signal datapaths. Its purpose is to make the analog/RF system **programmable, calibratable, testable, and suitable for practical silicon operation**.

3.1 Host Interface

An external **MCU, FPGA, or SDR host processor** communicates with the RFIC through three principal interfaces.

SPI is the main configuration interface. It programs operating frequency, RX/TX gain, filter bandwidth, PLL settings, VCO tuning, ADC/DAC modes, calibration controls, and test functions.

I²S provides the digital I/Q datapath. During reception, digital I/Q samples travel from the RFIC to the host. During transmission, I/Q samples travel from the host into the RFIC.

GPIO/IRQ signals provide hardware-level control and status functions such as interrupts, RF switch control, PA enable, calibration completion, and other real-time signals.

3.2 Register Bank and Sequencer

SPI transactions are stored in the **register bank**, creating a controlled interface between external software and internal hardware.

The **control sequencer** converts these register settings into correctly ordered chip operations. This is particularly important for analog and RF circuitry because blocks cannot always be enabled simultaneously.

A typical startup sequence may involve:

POR → Bias Enable → Reference Clock → PLL Enable → VCO Calibration → PLL Lock → RX/TX Enable

Similarly, the transmitter driver should not be enabled before its bias, LO, DAC, and baseband paths are in valid operating states.

The sequencer therefore provides deterministic transitions between modes such as sleep, standby, receive, transmit, calibration, and test.

3.3 Calibration Engine

Integrated CMOS analog parameters change with **process, supply voltage, temperature, mismatch, and package parasitics**. The architecture therefore includes calibration as a fundamental system function rather than treating it as an optional feature.

VCO band calibration searches the coarse tuning range of the oscillator. The controller selects a VCO tuning code, measures the resulting frequency, compares it with the required value, and iterates until an appropriate operating band is found.

RC calibration compensates integrated resistor and capacitor variation. The resulting trim values can be applied to RX and TX filters so their bandwidth remains close to the desired value across manufacturing and temperature variation.

IQ calibration corrects gain and phase differences between the I and Q signal paths. These mismatches are particularly important because they directly determine image rejection.

TX DC calibration compensates DC offsets in the DAC, baseband, and mixer paths. In a zero-IF transmitter, these offsets can appear as unwanted carrier leakage at the RF output.

Loopback modes provide controlled internal paths between TX, RX, and digital blocks. They simplify silicon bring-up, production testing, calibration, and fault isolation.

3.4 RX and TX Digital Processing

The RX DSP accepts high-rate sigma-delta ADC outputs and performs the required digital filtering and decimation. A representative processing sequence is:

$\Sigma\Delta$ ADC $\rightarrow$ CIC/Sinc Filtering $\rightarrow$ Halfband Filtering $\rightarrow$ DC Correction $\rightarrow$ IQ Correction $\rightarrow$ I²S

The TX DSP operates in the opposite direction:

I²S $\rightarrow$ Interpolation $\rightarrow$ Digital Filtering $\rightarrow$ $\Sigma\Delta$ Modulation $\rightarrow$ FIR DAC

These digital blocks allow several impairments of the analog signal chain to be corrected or relaxed, reducing the burden on transistor-level analog accuracy.

3.5 Power, Bias and POR

The power subsystem supports the entire architecture through separate supply domains. The planned organization includes dedicated supplies for **RF circuitry, VCO/synthesizer circuitry, analog baseband, digital logic, and 3.3 V I/O.**

Separating these domains is particularly important because digital switching noise can otherwise couple into sensitive RF and analog circuits. VCO supply noise may become phase noise, while noise coupled into the LNA or ADC can directly degrade receiver sensitivity.

The **bias subsystem** generates the reference currents and voltages required by the LNA, mixers, PLL, VCO, filters, ADCs, DACs, and RF driver. The **power-on-reset circuit** ensures that the RFIC begins operation in a known state and that digital control remains reset until the required supplies and references are valid.

3.6 System-Level Interpretation

The three architecture diagrams should ultimately be viewed together.

The **first diagram** defines the chip hierarchy and identifies the major design domains. The **second diagram** shows how information physically moves through the RFIC from RF to digital in receive mode and from digital to RF in transmit mode. The **third diagram** explains how the host, control logic, calibration engine, and power infrastructure manage those signal paths.

The central architectural principle is that these blocks cannot be optimized independently. **PLL/VCO performance affects RX and TX; LNA gain and noise influence ADC requirements; DAC performance determines TX filtering requirements; calibration influences allowable analog variation; and power/substrate isolation affects nearly every sensitive block.**

Therefore, each macro should eventually have a formal design specification containing its **electrical interface, supply domain, input/output loading, gain or conversion requirement, noise/linearity target, power budget, area budget, calibration hooks, PVT**

corners, Monte-Carlo requirements, and post-layout CACE pass/fail criteria. These macro specifications should remain subordinate to the top-level system architecture so that local design choices do not compromise overall RFIC performance.